

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0659

Firstenergy R Paul Smith Power Station 138 kV
August 16, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0659. The Interconnection Customer proposes a 57.0 MW solar photovoltaic facility interconnecting at the Firstenergy R Paul Smith Power Station 138 kV substation in the Maryland area. The request was received on August 04, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 215 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,905,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Frederick Substation – Altavista Power Station 138.0 kV Line
Contingency: Carl Bailey – Otsego 138.0 kV
Loading: 168.4% (Rating: 224 MVA)
DFAX: 5.3%
Pre-project loading: 167.0%
2. Facility: Lamont – Frederick 138.0 kV
Contingency: Carl Bailey – Otsego 138.0 kV
Loading: 143.4% (Rating: 288 MVA)
DFAX: 5.3%
Pre-project loading: 142.4%
3. Facility: Innerarity – Maryland Solar 138.0 kV
Contingency: Pineville – East Hagerstown 138.0 kV
Loading: 106.7% (Rating: 388 MVA)
DFAX: 6.4%
Pre-project loading: 105.8%
4. Facility: Pluess – Cameron 138.0 kV
Contingency: Kearneyville – Clark County 345.0 kV line
Loading: 106.3% (Rating: 368 MVA)
DFAX: 6.9%
Pre-project loading: 105.2%
5. Facility: Metro Center – Hollister 138.0 kV
Contingency: Pineville Substation – East Hagerstown 138.0 kV Line
Loading: 105.2% (Rating: 352 MVA)
DFAX: 6.4%
Pre-project loading: 104.1%

6. Facility: Firstenergy R Paul Smith Power Sta Substation – Dam No. 4 Hydro Station 138.0 kV
 Contingency: Lascassas – Dam No. 5 Hydro Station 138.0 kV L
 Loading: 103.2% (Rating: 264 MVA)
 DFAX: 40.4%
 Pre-project loading: 94.5%
7. Facility: Bellsfork – El Sobrante 138.0 kV
 Contingency: E Hagerstown – Pineville 138.0 kV line
 Loading: 101.2% (Rating: 403 MVA)
 DFAX: 6.4%
 Pre-project loading: 100.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Frederick – Altavista Power Sta 138.0 kV	Rebuild line with higher-capacity conductor	\$9,605,000	20% (\$1,918,000)
Lamont – Frederick 138.0 kV	Rebuild line with higher-capacity conductor	\$8,652,000	20% (\$1,728,000)
Innerarity – Maryland Solar 138.0 kV	Reconductor line	\$428,000	2% (\$10,000)
Pluess – Cameron 138.0 kV	Reconductor line	\$1,550,000	3% (\$41,000)
Hollister – Metro Ctr 138.0 kV Line	Reconductor line	\$1,162,000	2% (\$27,000)
Firstenergy R Paul Smith Power Sta Substation – Dam No. 4 Hydro Station 138.0 kV	Reconductor line	\$924,000	15% (\$142,000)
Bellsfork – El Sobrante 138.0 kV	Reconductor line	\$1,648,000	2% (\$39,000)
	Total allocated to project		\$3,905,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.